

Alzheimer's

Aricept= Donepezil *AChE inhib*

Namenda= Memantine

NMDA antag

Neuropathic Pain

Cymbalta= Duloxetine

Neurontin= Gabapentin

Lyrica= Pregabalin

Opioids

Full agonist:

Fentanyl

Methadone

Morphine

Heroin

Weak agonist:

Butrans= Buprenorphine (partial)

Hydrocodone

Lortab, Norco, Vicodin= Hydro/acet.

Dilaudid= Hydromorphone

OxyContin= Oxycodone

Percocet= Oxycodone/acet.

Codeine

Imodium= Loperamide

Demerol= Meperidine

Ultram= Tramadol

Drug + EtOH Use Tx: *antag*

Suboxone= Buprenorphine/Naloxone

Narcan= Naloxone (IM, nasal)

Vivitrol= Naltrexone (PO, IM)

Campriil= Acamprosate

Antabuse= Disulfiram

TCA's

Amitriptyline

Nortriptyline

Doxepin

Clomipramine

Desipramine

Imipramine

*inhibit
NE & 5-HT
reuptake*

MAOI's

Phenelzine

Selegiline

Tranylcypromine

SSRI's

Celexa= Citalopram

Lexapro= Escitalopram

Luvox= Fluvoxamine

Prozac= Fluoxetine

Paxil= Paroxetine

Zoloft= Sertraline

SNRI's

Cymbalta= Duloxetine

Milnacipran

Effexor= Venlafaxine

Pristiq= Desvenlafaxine

Serotonin Modulators

Viibryd= Vilazodone

Trintellix= Vortioxetine

Atypical Antidepressants

Wellbutrin= Bupropion

Remeron= Mirtazapine

Other Anti-Anxiety Drug

BuSpar= Buspirone

Vistaril= Hydroxyzine (PO, IM)

BENZOS

Xanax= Alprazolam

Klonopin= Clonazepam

Valium= Diazepam (PO, IM)

Ativan= Lorazepam (PO, IM)

Oxazepam

Restoril= Temazepam

Librium= Chlordiazepoxide

Benzo Overdose Tx:

Flumazenil (IV)

Bipolar Disorder: maintenance

*Lithium

***Depakote**= Valproate

***Lamictal**= Lamotrigine

Tegretol= Carbamazepine

Trileptal= Oxcarbazepine

inhibits Nav

GABA agonist

inhibits Nav, ↓ glut. release

inhibits Nav

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Typical Antipsychotic

Thorazine= Chlorpromazine

Haldol= Haloperidol (PO, IM)

Prolixin= Fluphenazine (PO, IM)

Atypical Antipsychotic

~**Abilify** (PO, IM), **Aristada** (IM)= Aripip

Rexulti= Brexpiprazole

Saphris= Asenapine

Clozaril= Clozapine

~**Zyprexa**= Olanzapine (PO, IM)

~**Symbyax**= Olanzapine/fluoxetine

***Seroquel**= Quetiapine

Vraylar= Cariprazine

~**Latuda**= Lurasidone

Invega= Paliperidone (PO, IM)

~**Risperdal**= Risperidone (PO, IM)

Geodon= Ziprasidone

EPS Tx:

Acute Dystonia: **Benzotropine**,

***Diphenhydramine**

Akathisia: **Benzotropine**, ***Propranolol**,

~**Lorazepam**

Parkinsonism: **Benzotropine**,

***Amantadine**

Tardive: **Valbenazine**,

Deutetrabenazine

ADHD Drugs

Stimulants: *blocks NE & D reuptake*

Mydayis, Adderall= Amph/dextroamp

Vyvanse= Lisdexamfetamine

Ritalin, Concerta= Methylphenidate

Non Stimulants:

Strattera= **Atomoxetine** *inhibits NE reuptake*

Kapvay= **Clonidine** *α₂ agonist*

Intuniv= Guanfacine

Narcolepsy Drugs

Nuvigil= Armodafinil

Provigil= Modafinil

Xyrem= Sodium Oxybate

Insomnia

Unisom= Diphenhydramine

Lunesta= Eszopiclone

Trazodone

Ambien= Zolpidem

PTSD nightmares

Prazosin

Doxazosin

RLS

Mirapex= Pramipexole

Requip= Ropinirole

Review Document-Psych

Antidepressants

SSRI

Fluoxetine	Prozac	Long ½ life, stimulating
Sertraline	Zoloft	GI Side effect
Paroxetine	Paxil	No Pregnant, QHS-sedating
Citalopram	Celexa	QTC prolong
Escitalopram	Lexapro	QTC prolong, less SE
Fluvoxamine	Luvox	OCD approved, lots of CYP interactions, don't use

Other Antidepressants

Bupropion	(Wellbutrin), NDRI(NE,DA), stimulating, no wt gain, smoking cessation, not for sz disorder or eating disorders, low Sexual SE, adjunct in depression too, XL best
Duloxetine	(cymbalta)SNR(5HT, NE)I, increases BP, neuropathic pain, fibromyalgia
Venlafaxine	(Effexor)SNR(5HT, NE)I, increases BP, neuropathic pain, fibromyalgia, withdrawal SE, needs titration
Mirtazapine	(Remeron) alpha 2antag, NE/5HT increase, wt gain, sedating (greater at low dose), good for cancer/cachexia
Trazodone	Priapism, for insomnia not anti-depress
Nefazodone (serzone)	Newer, not tested
Vilazodone (Viibryd)	
TCA: Amitriptyline Nortriptyline	(elavil), Anti HAM-(hist, adrenergic, muscarinic)--> tired orthostatic and dry Convulsions, coma, cardiotoxicity

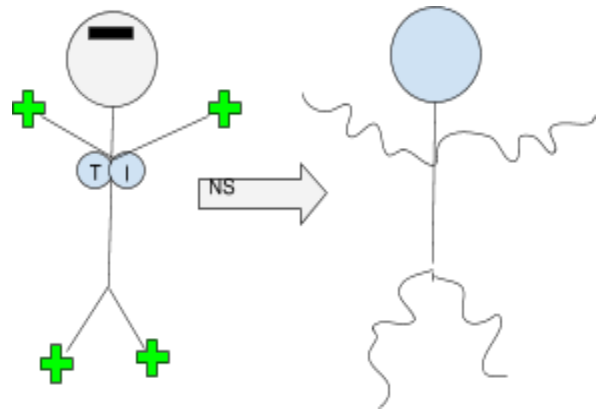
Dopamine Pathways:

Mesolimbic (+)

Mesocortical (-)

Tuberoinfundibular (TI) (prolactin/galactorrhea)

Nigrostriatal (NS) (EPS - Acute dystonia, Akathisia, parkinsonism, Tardive dyskinesia)



Antipsychotics

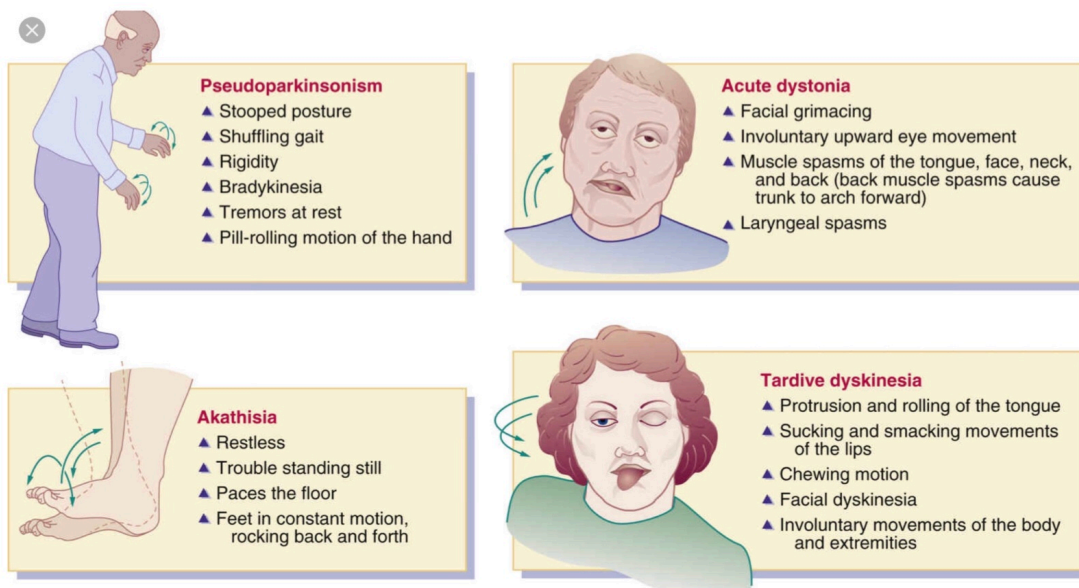
Clozapine	Clozaril	Agranulocytosis, blood draws, No D2 blockade, mostly D4, For refractory Schizo
Haloperidol	Haldol	1st gen, high potency, more EPS/NMS, less anti HAM, LAI
Thioridazine	Thorazine	1st gen, low potency, more anti HAM, less EPS/NMS
Risperidone	Risperdal	High D2 block, increase prolactin, LAI
Olanzapine	Zyprexa	Sedating, metabolic SE, IM, PO,
Ziprasidone	Geodon	Prolong QTC, fewest Metabolic SE, BID dosing, IM PO
Quetiapine	Seroquel	Sedating, requires high dose, lowest D2 block, used in parkinson's psychosis,
Aripiprazole	Abilify	Fewer SE, akathisia, LAI
Lurasidone	Latuda	Bipolar depression, Take with food, ("latud-a with food-a")

Mood Stabilizers (bipolar and seizures)

Lithium	TSH, Pregnancy, CMP, Ebstein, acne, Nephro DI, tremor, Increase PLT, prob ok in preg
Valproic Acid (depakote)	Neural tube defect, Hepatotoxic, hair loss, wt gain, PCOS?, Pancreatitis, NOT ok in Preg, used in agitation sometimes
Carbamazepine Oxcarbazepine	Autoinducer, Birthcontrol pills not working anymore,
Lamotrigine (Lamictal)	Stephen johnson, must titrate slowly, not useful in acute hospital situation
Topiramate	(topamax=dopamax) cognitive slowing, nephrolithiasis

Antipsychotic extrapyramidal effects		Pharmacotherapy*
Acute dystonia	Sudden, sustained contraction of the neck, mouth, tongue & eye muscles	- Benztropine - Diphenhydramine ★
Akathisia	Subjective restlessness, inability to sit still	- Beta blocker (<u>propranolol</u>) ★ - Benzodiazepine (<u>lorazepam</u>) - Benzotropine
Parkinsonism	Gradual-onset tremor, rigidity & bradykinesia	- Benzotropine - Amantadine
Tardive dyskinesia	Gradual onset after prolonged therapy (>6 months): dyskinesia of the mouth, face, trunk & extremities	- Valbenazine - Deutetrabenazine

*Management may include reducing the dose or switching to another antipsychotic, depending on the clinical scenario.



Medications, Electroconvulsive Therapy, and Side Effects

Type of medication	Adverse effects
Tricyclic antidepressants (amitriptyline, nortriptyline, imipramine)	^{ortho} Hypo/hypertension, dry mouth, constipation, confusion, <u>arrhythmias</u> , sexual side effects, weight gain, GI disturbances
Monoamine oxidase inhibitors (phenelzine, isocarboxazid, tranylcypromine)	Monitor diet, given that food rich in tyramine will produce hypertension. Safe foods include white wine and processed cheese. Unsafe foods include red wine, aged cheese, and chocolate.
Serotonin selective reuptake inhibitors (fluoxetine, paroxetine, sertraline, citalopram, escitalopram, fluvoxamine)	Headaches, weight changes, sexual side effects, GI disturbances
Serotonin norepinephrine reuptake inhibitors (venlafaxine, duloxetine, desvenlafaxine)	Hypertension, blurry vision, weight changes, sexual side effects, GI disturbances
Others (bupropion, mirtazapine, trazodone)	Bupropion has increased risk for <u>seizures</u> , trazodone has increased risk for <u>priapism</u> , and mirtazapine has increased risk for <u>weight gain</u> and sedation.
Lithium	Tremors, weight gain, GI disturbance, <u>nephrotoxic</u> , <u>teratogenic</u> , leukocytosis, <u>diabetes insipidus</u> . Severe toxicity gives confusion, ataxia, lethargy, and abnormal reflexes.
Valproic acid	Tremors, weight gain, GI disturbances, alopecia, <u>teratogenic</u> , <u>hepatotoxic</u> . Must monitor levels; toxicity causes hyponatremia, coma, or death.
Lamotrigine	Stevens-Johnson syndrome
Electroconvulsive therapy (ECT)	Headaches, transient memory loss

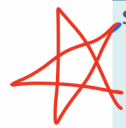
Adverse Effects of Atypical Antipsychotic Medications

Antipsychotic medication	Specific adverse effects
Olanzapine	Greater incidence of <u>diabetes</u> and <u>weight gain</u> ; avoid in <u>diabetic and obese patients</u>
Risperidone	Greater incidence of <u>movement disorders</u>
Quetiapine	<u>Lower incidence of movement disorders</u> ; <u>appropriate for use in patients with existing movement disorders</u>
Ziprasidone	Increased risk of <u>prolongation of QT interval</u> ; avoid in <u>patients with conduction defects</u>
Clozapine	High risk of <u>agranulocytosis</u> ; need to <u>monitor CBC</u> on regular basis; never used as first-line treatment given side-effect profile
Aripiprazole	Partial dopamine agonist, approved as adjunct treatment for major depressive disorder
Lurasidone	The only antipsychotic in Pregnancy Category B; <u>safer for use in pregnant patients</u>

NMS vs Serotonin Syndrome

	NMS	Serotonin Syndrome
Symptoms	-Rigidity*** -Fever*** -Confusion/AMS -Autonomic instability -Elevated CK	-Tremor*** -Diarrhea*** -Clonus** -Confusion/AMS -Flushing -Diaphoresis -Autonomic instability
Onset of Sx	-Anytime, more likely newly introduced Anti Psy within 1-3 days	-Minutes to hours
Duration	-1-2 weeks, if LAI 4-6 weeks	-Hours to days until resolution
Management	-ICU emergency*** -IVF -Dantrolene -Bromocriptine -ECT -Should have some response to fever/rigidity within hours, AMS resolves later	-Medical floor, maybe ICU -Cyproheptadine, 4mg, Antihistaminergic, 5HT antagonist, Oral medication -Benzos
Re-Introduction	-Re-introduce antipsychotics 2-4wks later -Use less Dopa blockade: Clozaril < Ziprasidone < Quetiapine < Abilify < olanzapine < invega < Risperdal < haldol	
Offending agents	-Antipsychotics 1st gen > 2nd gen Haldol > Thorazine -IM > IV > PO (IM more due to depo/muscle damage elevating ck) -Metoclopramide (anti emetic in cancer) -Withdrawal of parkinsonian drugs -Promethazine -Lithium -Dehydration with vigorous exercise on above agents	-Linezolid(MAOI) -SSRI -Bupropion -TCA (nortryp, amytrip) - MOAI -Tryptans -Diet drugs, phen.... -Flexeril -Trazodone -Tramadol -Lithium -2nd Gen Antipsychotics

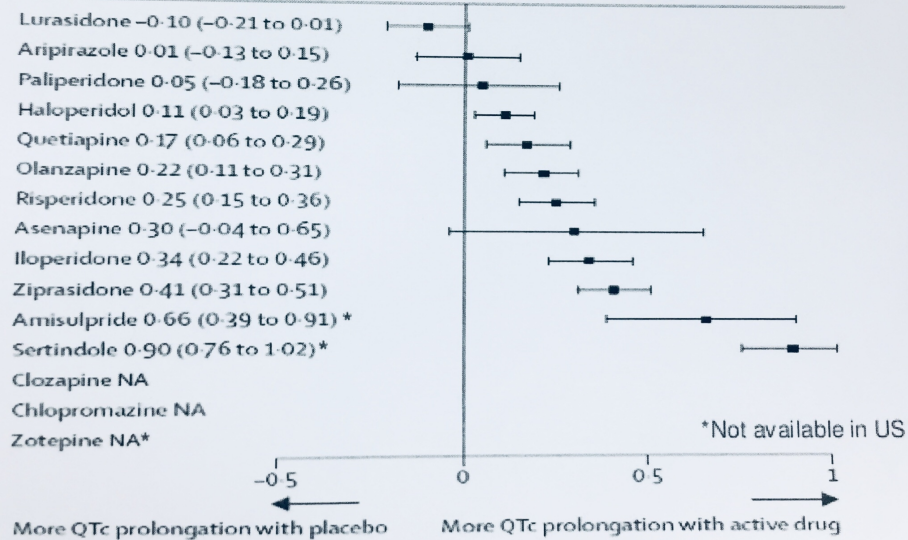
Psychiatric emergencies



	CAUSE	MANIFESTATION	TREATMENT
Serotonin syndrome	Any drug that ↑ 5-HT. Psychiatric drugs: MAO inhibitors, SSRIs, SNRIs, TCAs, vilazodone, vortioxetine Nonpsychiatric drugs: tramadol, ondansetron, triptans, linezolid, <u>MDMA</u> , dextromethorphan, meperidine, St. John's wort	3 A's: ↑ Activity (neuromuscular) Autonomic stimulation <u>Agitation</u> Symptoms of neuromuscular hyperactivity include <u>clonus</u> , <u>hyperreflexia</u> , <u>hypertonia</u> , <u>tremor</u> , <u>seizure</u> Symptoms of autonomic stimulation include <u>hyperthermia</u> , <u>diaphoresis</u> , <u>diarrhea</u> <i>high fever</i>	Cyproheptadine (5-HT ₂ receptor antagonist) <i>coma death</i> <i>ataxia</i> <i>↑CPK</i> <i>↑WBC's</i>
Carcinoid syndrome^a	Carcinoid tumor of GI tract, lung	Diarrhea, flushing, wheezing, right heart disease (if tumor is in the gut)	Octreotide

QTc prolongation

E QTc prolongation OR (95% CrI)



Psychiatric emergencies (continued)

	CAUSE	MANIFESTATION	TREATMENT
 Hypertensive crisis	Eating tyramine-rich foods (eg, aged <u>cheeses</u> , cured meats, wine) while taking MAO inhibitor	Hypertensive crisis (tyramine displaces other neurotransmitters [eg, NE] in the synaptic cleft → ↑ sympathetic stimulation)	<u>Phentolamine</u>
 Neuroleptic malignant syndrome	Antipsychotics + genetic predisposition ↑BP ↑HR	Malignant FEVER: Myoglobinuria Fever high Encephalopathy Vitals unstable ↑ Enzymes (eg, ↑ CK) Rigidity of muscles ("lead pipe")	<u>Dantrolene</u> , dopamine agonist (eg, <u>bromocriptine</u>), discontinue causative agent hyperK + ↑WBC's + metabolic acidosis
Malignant hyperthermia^a	Inhaled <u>anesthetics</u> , <u>succinylcholine</u> + genetic predisposition	Fever, severe muscle contractions	<u>Dantrolene</u>
Delirium tremens	Alcohol withdrawal; occurs 2–4 days after last drink Classically seen in hospital setting when inpatient cannot drink	Altered mental status (eg, <u>hallucinations</u>), autonomic hyperactivity, <u>anxiety</u> , <u>seizures</u> , <u>tremors</u> , psychomotor <u>agitation</u> , <u>insomnia</u> , <u>nausea</u>	Benzodiazepines (eg, <u>chlordiazepoxide</u> , <u>lorazepam</u> , <u>diazepam</u>)
Acute dystonia	<u>Typical</u> antipsychotics, anticonvulsants (eg, carbamazepine), metoclopramide	Sudden onset of muscle spasm, stiffness, oculogyric crisis that occurs within hours to days after medication use; can lead to laryngospasm requiring intubation	<u>Benztropine</u> or <u>diphenhydramine</u>
 Lithium toxicity	Change in lithium dosage or health status (narrow therapeutic window), <u>concurrent use of thiazides</u> , <u>ACE inhibitors</u> , <u>NSAIDs</u> , or other nephrotoxic agents	<u>Nausea</u> , <u>vomiting</u> , slurred speech, <u>hyperreflexia</u> , <u>seizures</u> , <u>ataxia</u> , nephrogenic <u>diabetes insipidus</u> tremors	Discontinue lithium, hydrate aggressively with <u>isotonic sodium chloride</u> , consider hemodialysis get EKG
 Tricyclic antidepressant toxicity	TCA overdose ↓BP (orthostatic) ↑HR dilated pupils	Respiratory depression, <u>hyperpyrexia</u> , <u>prolonged QT interval</u> Tri-C's: <u>Convulsions</u> <u>Coma</u> <u>Cardiotoxicity</u> (arrhythmia due to Na ⁺ channel inhibition)	Supportive treatment, monitor ECG, <u>NaHCO₃</u> (prevents arrhythmia), activated charcoal get EKG

^aCarcinoid syndrome and malignant hyperthermia are not psychiatric emergencies, but are included for comparison with serotonin syndrome and neuroleptic malignant syndrome, respectively.

Teratogens

Most susceptible in 3rd–8th weeks (embryonic period—organogenesis) of pregnancy. Before week 3, “all-or-none” effects. After week 8, growth and function affected.

TERATOGEN	EFFECTS ON FETUS	NOTES
Medications		
ACE inhibitors	Renal damage	
Alkylating agents	Absence of digits, multiple anomalies	
Aminoglycosides	Ototoxicity	A mean guy hit the baby in the ear.
Antiepileptic drugs	Neural tube defects, cardiac defects, cleft palate, skeletal abnormalities (eg, phalanx/nail hypoplasia, facial dysmorphism)	High-dose folate supplementation recommended. Most commonly valproate, carbamazepine, phenytoin, phenobarbital.
Diethylstilbestrol	Vaginal clear cell adenocarcinoma, congenital Müllerian anomalies	
Folate antagonists	Neural tube defects	Includes trimethoprim, methotrexate, antiepileptic drugs.
Isotretinoin	Multiple severe birth defects	Contraception mandatory. Iso TERAT inoin.
Lithium	Ebstein anomaly (apical displacement of tricuspid valve)	
Methimazole	Aplasia cutis congenita	
Tetracyclines	Discolored teeth, inhibited bone growth	“Teethracyclines.”
Thalidomide	Limb defects (phocomelia, micromelia—“flipper” limbs)	Limb defects with “tha-limb-domide.”
Warfarin	Bone deformities, fetal hemorrhage, abortion, ophthalmologic abnormalities	Do not wage warfare on the baby; keep it heppy with heparin (does not cross placenta).
Substance abuse		
Alcohol	Common cause of birth defects and intellectual disability; fetal alcohol syndrome	
Cocaine	Low birth weight, preterm birth, IUGR, placental abruption	Cocaine → vasoconstriction.
Smoking (nicotine, CO)	Low birth weight (leading cause in developed countries), preterm labor, placental problems, IUGR, SIDS, ADHD	Nicotine → vasoconstriction. CO → impaired O ₂ delivery.
Other		
Iodine (lack or excess)	Congenital goiter or hypothyroidism (cretinism)	
Maternal diabetes	Caudal regression syndrome (anal atresia to sirenomelia), congenital heart defects (eg, VSD, transposition of the great vessels), neural tube defects, macrosomia, neonatal hypoglycemia, polycythemia	
Methylmercury	Neurotoxicity	Highest in swordfish, shark, tilefish, king mackerel.
Vitamin A excess	Extremely high risk for spontaneous abortions and birth defects (cleft palate, cardiac)	
X-rays	Microcephaly, intellectual disability	Minimized by lead shielding.